

The Pulickal Conjecture

Finite-Range Extremal Complexity of Signed Pure Prime-Power Representations

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Abstract

We study a finite, exact extremal problem for signed representations of integers by *pure powers of finitely many fixed primes*. The basic resource is the number of atomic summands, not the numerical size of an unrestricted coefficient. To separate target size from the permitted size of the building blocks, we introduce a two-parameter model. For a target bound X and an atom-height bound H , let $\lambda_{k,H}(n)$ be the minimum number of signed atoms needed to represent n , where the atoms are 1 and the positive powers of the first k primes not exceeding H . We then define

$$M_k(X, H) = \max_{\substack{p \leq X \\ p \text{ prime}}} \lambda_{k,H}(p).$$

The original one-parameter quantity is the diagonal specialization $M_k(B) = M_k(B, B)$.

The two-parameter formulation exposes a distinction that is easy to blur in informal discussions. A finite-height problem is algorithmically finite, whereas the corresponding unrestricted word metric permits arbitrarily large powers and therefore admits cancellation phenomena absent from the bounded model. We establish elementary monotonicity properties, an exact coefficient-cost formulation, and a counting lower bound. In particular, when H is at most a fixed power of X , every fixed term budget covers only a polylogarithmic number of primes up to X , and consequently

$$M_k(X, H) \geq (1 - o(1)) \frac{\log X}{\log \log X}.$$

For the diagonal model this gives

$$M_k(B, B) \geq (1 - o(1)) \frac{\log B}{\log \log B},$$

while signed non-adjacent binary expansion gives the unconditional upper bound $M_k(B, B) = O(\log B)$.

We formulate the **Pulickal Conjecture**: for every fixed $k \geq 2$,

$$M_k(B, B) = \Theta_k \left(\frac{\log B}{\log \log B} \right).$$

The lower bound is proved; the conjectural content is the matching upper bound. We place this statement carefully relative to Nathanson's word-length and lambda-sequence framework, work of Hajdu and Tijdeman on linear combinations of powers and power products, results of Ádám–Hajdu–Luca on S -unit combinations, double-base representations, and the multibase Hamming-weight theorem of Krenn–Suppakitpaisarn–Wagner. None of those results, as stated, supplies the conjectured upper bound for the present *pure-power*, *prime-target*, *finite-height* model.

We also organize the computational side as a reproducibility protocol rather than as a substitute for proof. Exact bitset dynamic programming gives a finite certificate for each fixed (k, X, H) , and the paper records a representative data set for $k = 4$, $X = H = 10^7$. The numerical evidence is treated cautiously: it shows an economical finite regime, but by itself it cannot distinguish the conjectured scale from other slowly growing alternatives.

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1 Introduction

The elementary observation motivating this paper is that surprisingly large integers, including primes, can sometimes be assembled from a very small number of powers of a few small primes. For example,

$$5 = 3 + 2, \quad 7 = 3^2 - 2, \quad 11 = 3^3 - 2^4, \quad 13 = 3^2 + 2^2.$$

The observation becomes mathematically interesting only after the cost model is made explicit. If arbitrary integer coefficients are allowed for free, an expression such as

$$P = X3^m - Y2^n + Z5^\ell$$

can conceal essentially all of the information about P inside X, Y, Z . We therefore count each copy of an allowed atom as one unit of complexity. In coefficient form this is an ℓ^1 objective.

The resulting question belongs to a well-developed family of additive word-metric problems. Nathanson introduced arithmetic-diameter and lambda-sequence questions for infinite generating sets, and Hajdu and Tijdeman studied representations by linear combinations of powers of fixed integers. The broader theory also includes representations by S -units and by power products, as well as double-base and multibase digital systems. The contribution proposed here is deliberately narrower: it isolates a prime-target extremal statistic for the *union of pure powers* of a fixed finite prime palette, imposes a finite atom-height bound, and studies the resulting extremal complexity quantitatively.

The finite-height restriction is not merely a coding convenience. In an unrestricted signed representation, powers much larger than the target may occur in intermediate expressions and cancel. Consequently, a short unrestricted representation need not be visible inside the atom set bounded by the target. To make this distinction structural, we use two parameters throughout the revised paper:

- X , the size of the target range;
- H , the largest permitted atom.

The original model is then the diagonal case $H = X$.

The principal question is therefore:

For a fixed palette of at least two small prime bases, how many signed pure-prime-power atoms are needed, in the worst case over primes up to X , when atoms are restricted to height H ? In particular, what happens on the natural diagonal $H = X$?

The main conjecture predicts that, on the diagonal and for every fixed $k \geq 2$,

$$M_k(B, B) \asymp \frac{\log B}{\log \log B}.$$

The lower bound is elementary and comes from counting the available atoms and the number of short expressions. The upper bound is not proved here. Its difficulty lies in showing that the actual signed sumset of the sparse pure-power dictionary is sufficiently expansive to reach every prime target at essentially the counting threshold.

The paper makes four claims that should be kept logically separate:

1. the representation model is finite and exact for fixed (k, X, H) ;
2. the counting obstruction forces a term count of order at least $\log X / \log \log X$ when H is not excessively larger than X ;
3. signed binary arithmetic gives a general logarithmic upper bound;

4. the matching $O(\log X / \log \log X)$ upper bound for the pure-power prime-target model is conjectural.

This separation is important. The paper is not claiming a new theorem about general S -unit representations, and it is not claiming that the finite data prove an asymptotic law. Rather, it proposes a specific quantitative problem and develops a framework in which both theoretical and computational progress can be measured precisely.

2 Relation to the Existing Literature

2.1 Nathanson's word metric and lambda sequences

Let A be an infinite generating set for a group G . Nathanson considers the associated word metric and, in situations of infinite diameter, defines lambda sequences by asking for the smallest positive element whose word length is a prescribed value. For finite sets of primes, the generating set of positive integers supported on those primes leads to an infinite-diameter problem and a rich collection of extremal questions [1].

The present statistic is best viewed as a finite-range and target-restricted relative of this general word-metric viewpoint. We do not claim that the underlying idea of measuring representation length is new. The new emphasis is the particular diagonal finite-height statistic

$$M_k(B, B) = \max_{p \leq B} \lambda_{k,B}(p),$$

together with its two-parameter extension $M_k(X, H)$ and the computational study of the prime frontier.

2.2 Pure powers versus power products

The distinction between pure powers and power products is fundamental. In the present paper the allowed positive atoms are

$$1, \quad 2^a, 3^b, 5^c, \dots,$$

with one base occurring in each non-unit atom. In a power-product system, terms such as

$$2^a 3^b 5^c$$

are also allowed as single atoms. The latter system is much denser.

Hajdu and Tijdeman studied both sums of powers and sums of power products [2, 3]. Their power-product paper treats a set of all positive integers whose prime factors lie in a fixed finite set and studies the smallest integer that cannot be represented by a bounded number of such terms. These results establish a broad theoretical background for sparse additive representations.

The multibase theorem of Krenn, Suppakitpaisarn and Wagner is particularly relevant. For multiplicatively independent bases and a finite digit set, they prove that the minimal Hamming weight has order

$$\frac{\log n}{\log \log n},$$

using a nontrivial analysis of a greedy algorithm and a Diophantine-approximation input, with a counting lower bound in the opposite direction [6]. That theorem does not directly settle the present conjecture because a mixed product such as $2^a 3^b$ is one atom in their system but is not an atom here. The pure-power dictionary is strictly thinner.

This distinction should not be presented as a technical footnote. It is the central structural obstacle in the present problem. The paper asks whether the $\log n / \log \log n$ phenomenon survives after all mixed monomials are removed, at least for a fixed set of prime bases and prime targets.

2.3 S -units and sparsity of fixed-term representations

The theory of S -unit equations gives a complementary perspective. Ádám, Hajdu and Luca proved that, for a fixed finite set of coefficients and a fixed number of S -unit terms, the represented integers up to x form a very sparse set; they also deduced that almost all primes are outside such fixed-complexity representation sets. Their paper contains a more precise pure-power analysis and explicitly treats distributions of values represented by linear combinations of pure powers [4].

This literature is fully compatible with the present conjecture. The Pulickal Conjecture does not assert that a fixed number of terms represents all sufficiently large primes. On the contrary, the conjecture forces the number of terms to tend to infinity, at the near-minimal rate allowed by the elementary counting obstruction. The conceptual transition is therefore from fixed-term sparsity to growing-term coverage.

2.4 Double-base representations

The signed two-base system generated by 2 and 3 is especially well studied. Dimitrov and Howe define double-base representations using signed integers having no prime divisors outside $\{2, 3\}$ and obtain substantial lower bounds on shortest lengths, together with exact computational values for the first integers escaping small term budgets [5]. Their model allows mixed terms $\pm 2^a 3^b$, so it remains richer than the pure-power model considered here.

The comparison nevertheless gives an important methodological lesson: exact extremal computations can be mathematically meaningful when accompanied by certification of nonexistence. The present paper adopts the same standard in a different, thinner dictionary.

2.5 Prime-power additive conjectures with varying bases

Stricker’s 2025 preprint conjectures that every integer greater than 23 is a sum of at most five prime powers with exponent at least 2, where the prime base is allowed to vary with the target. The conjecture is supported by exhaustive computations through 10^7 and by additional sampling [8]. That question is superficially similar but structurally different. Here the set of bases is fixed once and for all, signed terms are allowed, the complexity counts atomic copies, and all atoms are subject to an explicit finite height bound.

2.6 What is and is not being claimed as novel

The safest formulation is the following. The general philosophy of shortest representations by powers is established. The exact two-parameter statistic $M_k(X, H)$ for prime targets, its diagonal specialization, and the particular computational frontier are proposed here as a focused object of study. Any future journal submission should repeat the literature search in MathSciNet, zbMATH, Google Scholar and sequence databases using all equivalent formulations, including “signed pure powers”, “linear combinations of fixed powers”, “word length”, “arithmetic diameter”, “lambda sequence”, and “prime-target sparse representations”.

3 The Two-Parameter Representation Model

3.1 Prime palettes

Let

$$p_1 = 2, \quad p_2 = 3, \quad p_3 = 5, \quad p_4 = 7, \quad p_5 = 11, \dots$$

be the increasing sequence of primes. For $k \geq 1$, put

$$S_k = \{p_1, \dots, p_k\}.$$

The hypothesis $k \geq 2$ is the natural regime in which several multiplicatively independent prime bases are present.

3.2 Positive atoms

For a height parameter $H \geq 2$, define

$$A_k(H) = \{1\} \cup \{p_i^a : 1 \leq i \leq k, a \geq 1, p_i^a \leq H\}.$$

The explicit separation of 1 from the genuine prime powers removes a minor terminological ambiguity: 1 is an allowed unit atom because it is convenient for padding exact-length representations, but it is not itself a prime power in the usual convention.

Repeated use of an atom is permitted. The signed dictionary is

$$\pm A_k(H) = \{\pm a : a \in A_k(H)\}.$$

3.3 Finite-height signed length

Definition 3.1. For $n \in \mathbb{Z}$ with $n \geq 1$, define

$$\lambda_{k,H}(n) = \min \left\{ \sum_{a \in A_k(H)} |c_a| : n = \sum_{a \in A_k(H)} c_a a, \quad c_a \in \mathbb{Z} \right\}.$$

Because $2 \in S_k$ for every $k \geq 1$, the value is finite. Equivalently, n may be written as a signed sum of r atoms, and the minimum possible r is $\lambda_{k,H}(n)$.

The coefficient and atomic formulations are exactly equivalent: a coefficient $c_a > 0$ means c_a positive copies of a , while $c_a < 0$ means $|c_a|$ negative copies.

Remark 3.2. *The presence of the unit atom 1 implies that a representation using at most r terms can always be padded to one using exactly $r + 2s$ terms by inserting $1 - 1$ pairs. Thus dynamic programming over exact term layers can recover the usual minimum over “at most r ” terms without ambiguity.*

3.4 Target-range extremal functions

For $X \geq 2$ and $H \geq 2$, define the prime extremal function

$$M_k(X, H) = \max_{\substack{p \leq X \\ p \text{ prime}}} \lambda_{k,H}(p).$$

We also define the all-integer analogue

$$N_k(X, H) = \max_{1 \leq n \leq X} \lambda_{k,H}(n).$$

The original one-parameter statistic is the diagonal

$$M_k(B) := M_k(B, B), \quad N_k(B) := N_k(B, B).$$

The two-parameter notation is useful for three reasons. First, it makes the role of the height cutoff explicit. Second, it separates finite-range mathematics from the unrestricted problem. Third, it allows one to ask whether polynomially larger or smaller height budgets alter the conjectured scale.

3.5 Unrestricted length

Let

$$\lambda_k(n) = \inf_{H \geq 2} \lambda_{k,H}(n).$$

Because any particular finite representation uses only finitely many atoms, the infimum is in fact attained and may be regarded as a minimum over unrestricted finite representations. Moreover,

$$\lambda_k(n) \leq \lambda_{k,H}(n)$$

for every H .

Define

$$M_k^\infty(X) = \max_{\substack{p \leq X \\ p \text{ prime}}} \lambda_k(p).$$

Then

$$M_k^\infty(X) \leq M_k(X, H)$$

for every H , and in particular

$$M_k^\infty(B) \leq M_k(B, B).$$

The gap between these two quantities is a genuine subject of study rather than a notational nuisance.

4 Basic Structural Properties

The two-parameter notation makes several elementary monotonicities transparent.

Proposition 4.1 (Monotonicity in height). *For fixed k and n , if $H_1 \leq H_2$, then*

$$\lambda_{k, H_2}(n) \leq \lambda_{k, H_1}(n).$$

Consequently, for fixed k and X ,

$$M_k(X, H_2) \leq M_k(X, H_1).$$

Proof. Increasing the height enlarges the atom dictionary and therefore can only add admissible representations. Taking maxima over the same target set preserves the inequality. $\square$

Proposition 4.2 (Monotonicity in the palette). *For fixed X, H and $k \geq 1$,*

$$\lambda_{k+1, H}(n) \leq \lambda_{k, H}(n), \quad M_{k+1}(X, H) \leq M_k(X, H).$$

Proof. The $(k+1)$ -st palette contains every atom available to the k -th palette and possibly additional atoms. $\square$

Remark 4.3 (The diagonal extremum need not be monotone). *For a fixed target n , increasing B can only decrease $\lambda_{k, B}(n)$. However, $M_k(B, B)$ takes a maximum over a larger set of prime targets as B increases. An old hard prime may become easier while a newly admitted prime may be harder. Therefore no monotonicity of the diagonal extremal sequence should be assumed.*

4.1 Counting the atom dictionary

For fixed k ,

$$|A_k(H)| = 1 + \sum_{i=1}^k \left\lfloor \frac{\log H}{\log p_i} \right\rfloor = C_k \log H + O_k(1),$$

where

$$C_k = \sum_{i=1}^k \frac{1}{\log p_i}.$$

The formula follows directly from counting the exponents of each fixed base.

The key feature is not the exact value of C_k , but the order $|A_k(H)| \asymp_k \log H$. The pure-power dictionary is therefore logarithmically sparse in the height parameter.

5 The Counting Barrier

5.1 A finite-range counting proposition

The following proposition is the basic lower-bound engine of the paper.

Proposition 5.1 (Counting lower bound with separate target and height scales). *Fix $k \geq 1$. Suppose $X \rightarrow \infty$ and $H = H(X) \rightarrow \infty$ satisfy*

$$\log H \leq (\log X)^A$$

for some fixed constant $A > 0$; in particular, any polynomial height bound $H \leq X^\alpha$ is allowed. Then for every $\varepsilon > 0$ and all sufficiently large X ,

$$M_k(X, H) \geq (1 - \varepsilon) \frac{\log X}{\log(2|A_k(H)|)}.$$

In particular, if $H \leq X^\alpha$ for fixed $\alpha > 0$, then

$$M_k(X, H) \geq (1 - o(1)) \frac{\log X}{\log \log X}.$$

Proof. Let $m = |A_k(H)|$. There are at most $2m$ signed atoms. The number of ordered signed expressions of lengths $0, 1, \dots, r$ is at most

$$1 + (2m) + (2m)^2 + \dots + (2m)^r \leq (2m)^{r+1}.$$

If every prime $p \leq X$ had length at most r , then necessarily

$$\pi(X) \leq (2m)^{r+1}.$$

Taking logarithms gives

$$\log \pi(X) \leq (r + 1) \log(2m).$$

Since $\log \pi(X) = (1 + o(1)) \log X$ by the prime number theorem, choosing

$$r = \left\lfloor (1 - \varepsilon) \frac{\log X}{\log(2m)} \right\rfloor$$

produces a contradiction for sufficiently large X . This yields the first inequality.

If $H \leq X^\alpha$, then

$$m = C_k \log H + O_k(1) = O_k(\log X),$$

so

$$\log(2m) = \log \log X + O_{k,\alpha}(1).$$

Substitution gives the second assertion. □

5.2 The diagonal lower bound

Corollary 5.2 (Diagonal counting barrier). *For every fixed $k \geq 1$,*

$$M_k(B, B) \geq (1 - o(1)) \frac{\log B}{\log \log B}.$$

Equivalently,

$$\liminf_{B \rightarrow \infty} \frac{M_k(B, B) \log \log B}{\log B} \geq 1.$$

The proof is immediate from proposition 5.1 with $H = X = B$.

Remark 5.3 (Why the constant 1 is a counting constant). *The leading constant 1 in the lower bound is not claimed to be the conjectural leading constant of $M_k(B, B)$. It is the coefficient forced by the crude fact that there are only $O(\log B)$ available atoms. Collisions among different formal sums can only make coverage harder, so a future sharp upper bound could have a larger leading constant. The conjecture concerns order of magnitude, not the existence of an asymptotic constant.*

5.3 A stronger fixed-term consequence

The same counting argument yields a useful statement that directly interfaces with the fixed-term S -unit literature.

Proposition 5.4 (Fixed term budgets cover very few primes). *Fix $k, r \geq 1$. If $H \leq X^\alpha$ for some fixed $\alpha > 0$, then the number of primes $p \leq X$ satisfying*

$$\lambda_{k,H}(p) \leq r$$

is

$$O_{k,r,\alpha}((\log X)^r).$$

Consequently, the proportion of primes $p \leq X$ having $\lambda_{k,H}(p) \leq r$ tends to 0.

Proof. There are $m = O_k(\log X)$ available positive atoms and at most $2m$ signed atoms. The number of ordered expressions of length at most r is $O_r(m^r) = O_{k,r}((\log X)^r)$. Every prime with a representation of length at most r must occur among these values. Since $\pi(X) \sim X/\log X$, the resulting proportion tends to zero. $\square$

This proposition is deliberately elementary. It does not attempt to sharpen the much deeper S -unit counting theorems, but it isolates the exact obstruction relevant to the finite-height diagonal model.

6 The Unconditional Upper Bound

Because 2 belongs to every palette, every integer can be represented by signed binary arithmetic. A non-adjacent form gives a cleaner constant than ordinary binary expansion.

Proposition 6.1 (Signed-binary upper bound). *For every fixed $k \geq 1$ and every $B \geq 2$,*

$$M_k(B, B) \leq \frac{\log B}{2 \log 2} + O(1).$$

More generally,

$$N_k(B, B) \leq \frac{\log B}{2 \log 2} + O(1).$$

Proof. Every positive integer n has a non-adjacent signed binary expansion

$$n = \sum_j d_j 2^j, \quad d_j \in \{-1, 0, 1\},$$

with no two consecutive nonzero digits. If n has binary length L , then the number of nonzero digits is at most $(L+1)/2$. Every power 2^j that occurs is at most $n \leq B$, and hence is an admissible atom when $H = B$. Therefore

$$\lambda_{k,B}(n) \leq \frac{L+1}{2} \leq \frac{\log B}{2 \log 2} + O(1).$$

Taking the maximum over either primes or all integers proves the proposition. $\square$

Combining corollary 5.2 and proposition 6.1 gives

$$(1 - o(1)) \frac{\log B}{\log \log B} \leq M_k(B, B) \leq \frac{\log B}{2 \log 2} + O(1).$$

The lower and upper scales are different by a factor of order $\log \log B$.

7 The Pulickal Conjecture

Conjecture 7.1 (Pulickal Conjecture). *For every fixed integer $k \geq 2$,*

$$\boxed{M_k(B, B) = \Theta_k\left(\frac{\log B}{\log \log B}\right)} \quad (B \rightarrow \infty).$$

Equivalently, for every fixed $k \geq 2$, there are constants $0 < c_k \leq C_k < \infty$ such that

$$c_k \frac{\log B}{\log \log B} \leq M_k(B, B) \leq C_k \frac{\log B}{\log \log B}$$

for all sufficiently large B .

Because the lower bound is already proved, the substantive assertion is

$$M_k(B, B) = O_k\left(\frac{\log B}{\log \log B}\right).$$

The role of the threshold $k \geq 2$ is conceptual rather than formal in the lower bound. For $k = 1$, the system collapses to a single positional base and its natural worst-case behavior is logarithmic. The conjecture asserts that the introduction of at least two multiplicatively independent prime bases changes the extremal order.

7.1 Why the diagonal formulation is natural

The diagonal $H = X = B$ gives every target a supply of atoms whose sizes are commensurate with the target range. It is therefore the cleanest finite model in which to ask for worst-case prime coverage without allowing arbitrarily large cancelling terms.

A more general family of questions is

$$M_k(X, H),$$

with $H = X^\alpha$, $H = X$, $H = X^{1+o(1)}$, or other growth regimes. The counting lower bound remains on the same first-order scale whenever $\log H = O(\log X)$, while the unrestricted problem corresponds informally to letting $H \rightarrow \infty$.

8 A Precise Frontier Formulation

For fixed k, X, H and integer $h \geq 0$, define the prime deficit count

$$D_k(X, H; h) = \#\{p \leq X : p \text{ prime, } \lambda_{k,H}(p) > h\}.$$

Equivalently, define the represented-prime proportion

$$\rho_k(X, H; h) = 1 - \frac{D_k(X, H; h)}{\pi(X)}.$$

Then

$$M_k(X, H) \leq h \iff D_k(X, H; h) = 0 \iff \rho_k(X, H; h) = 1.$$

This formulation separates three notions that are sometimes conflated:

1. *density*: whether most primes are representable;
2. *coverage*: whether every prime in a finite range is representable;
3. *extremality*: the smallest term budget that gives complete coverage.

The main conjecture is an extremal statement. It is therefore strictly stronger than any density-one assertion at the same order.

8.1 First escaping prime

A finite analogue of a lambda frontier is

$$Q_k(X, H; h) = \min\{p \leq X : p \text{ prime and } \lambda_{k,H}(p) > h\},$$

with $Q_k(X, H; h) = +\infty$ if no escaping prime exists. Then

$$M_k(X, H) \leq h \iff Q_k(X, H; h) = +\infty.$$

As h increases, $Q_k(X, H; h)$ can only move to the right or disappear. Recording this frontier is often more informative computationally than recording only the terminal maximum.

8.2 Dual inversion

Ignoring lower-order effects, the conjectural relation

$$h \asymp \frac{\log B}{\log \log B}$$

formally inverts to

$$\log B \asymp h \log h, \quad B = h^{\Theta(h)}.$$

This is a superexponential frontier in the term parameter. The inversion is heuristic because the theorem concerns an extremal maximum and because constants may depend on k , but it is a useful way to interpret finite experiments: an observed record prime at length h should be compared against a scale of the form h^{ch} rather than merely e^{ch} .

9 Why the Conjectured Scale Is Plausible

9.1 The counting model

Let $m = |A_k(B)| \asymp \log B$. The number of coefficient vectors with ℓ^1 norm at most r is at most exponential in

$$r \log \left(\frac{m+r}{r} \right) + O(r),$$

which in the range $r \ll m$ behaves like

$$r \log \frac{m}{r} + O(r).$$

Setting this comparable to $\log B$ again suggests

$$r \asymp \frac{\log B}{\log \log B}.$$

This heuristic is not independent evidence in the probabilistic sense; it is a refined restatement of the same entropy barrier underlying the rigorous counting lower bound. Its value is that it identifies the expected transition scale at which a sparse dictionary could in principle have enough short combinations to cover an interval of exponential size.

9.2 What a matching upper bound would require

A proof of the conjecture cannot merely count formal expressions. It would have to control the arithmetic distribution of their values. In a useful idealization, one would need to show that sufficiently short signed sums from

$$\{2^a, 3^b, \dots, p_k^c\}$$

cover all primes in the target range, perhaps by proving a suitable additive expansion statement for iterated sumsets.

The difficulty is visible already at the local level. Pure powers have strong congruence structure, huge multiplicative gaps at large exponents, and many exact additive relations. A random-sum model therefore cannot be taken literally. The conjecture is plausible because several independent bases produce many more opportunities than a single positional system, but the mechanism converting this redundancy into uniform coverage remains unknown.

9.3 Why two bases might matter

For one base, every admissible atom lies on a single geometric progression. Signed digit systems then retain a positional rigidity that naturally leads to logarithmic worst-case lengths. With two multiplicatively independent bases, the sequences

$$2^a \quad \text{and} \quad 3^b$$

interlace in a genuinely nonperiodic manner. More generally, the logarithms $\log p_i$ are linearly independent over the rationals in the multiplicative sense relevant to the problem.

This interlacing should be regarded as a source of combinatorial redundancy, not as a proof mechanism. The richer power-product model exploits many mixed exponents simultaneously; the present pure-power model has far fewer atoms. Establishing that the remaining redundancy suffices at the $\log B / \log \log B$ scale is precisely the central challenge.

9.4 Reasons the conjecture could fail

There are several credible obstruction mechanisms:

1. **Local congruence obstructions.** Short sums may occupy only a small proportion of residue classes for carefully chosen moduli.
2. **Additive collisions.** A large number of formal expressions need not yield a comparably large set of distinct integers.
3. **Exceptional intervals.** Even if density tends to one for most targets, sparse regions could contain prime targets requiring more terms and determine the extremal maximum.
4. **Pure-power sparsity.** The dictionary is only logarithmic in the height, substantially thinner than power-product dictionaries.
5. **Prime-specific avoidance.** Primes may systematically avoid certain short-sum sets even when generic integers do not.

A successful proof must therefore be robust against both global sparsity and local arithmetic structure.

10 An Additive-Combinatorial Reformulation

Let

$$\pm A_k(H) = \{\pm a : a \in A_k(H)\}.$$

For $h \geq 0$, define the h -fold signed sumset

$$\Sigma_{k,H}(h) = \underbrace{\pm A_k(H) + \cdots + \pm A_k(H)}_{h \text{ summands}}, \quad \Sigma_{k,H}(0) = \{0\}.$$

Because $1 \in A_k(H)$, membership in a sufficiently large exact layer is equivalent to representability with a bounded number of terms.

The conjecture can then be restated as follows:

For every fixed $k \geq 2$, the prime points of the diagonal target interval $[1, B]$ are contained in $\Sigma_{k,B}(h)$ for

$$h \leq C_k \frac{\log B}{\log \log B}$$

for some constant C_k and all sufficiently large B .

This formulation suggests intermediate targets. Rather than proving complete coverage immediately, one can ask whether

$$\frac{|\Sigma_{k,B}(h) \cap [1, B] \cap \mathbb{P}|}{\pi(B)} \rightarrow 1$$

for

$$h = C_k \frac{\log B}{\log \log B}.$$

A density-one result would not prove the extremal conjecture, but it would show that the proposed scale is not merely an artifact of the worst case.

11 Exact Finite Computation

11.1 Why the computation is exact

Fix k, X, H . The atom set $A_k(H)$ is finite. For a term budget h , every exact h -term sum lies in

$$[-hH, hH]$$

because every atom has absolute value at most H . Define

$$E_0 = \{0\},$$

and recursively

$$E_{h+1} = E_h + (\pm A_k(H)).$$

Then E_h is exactly the set of integers representable as a sum of exactly h signed atoms.

Representing E_h by a bitset over $[-hH, hH]$ makes the recurrence exact: for each atom a , shift the bitset for E_h by $+a$ and by $-a$, then take the union over all atoms. No floating-point computation is involved.

11.2 Recovering minimal lengths

For every prime $p \leq X$, maintain an initially unset value. Whenever p first appears in E_h , record $\lambda_{k,H}(p) = h$. Because $1 \in A_k(H)$, a representation using at most h terms can be padded to one using exactly $h + 2s$ terms, so the first relevant layer correctly identifies the minimum.

The algorithm may stop as soon as every prime $p \leq X$ has been assigned a length. The maximum recorded length is then $M_k(X, H)$.

11.3 Certification protocol

An exact computation should be published with the following information:

1. source code for the bitset implementation;
2. a second implementation, preferably based on sparse sets, for small and medium bounds;
3. the complete machine-readable list of prime lengths or at least the hard-prime set;
4. a list of one shortest representation for every reported record prime;
5. arithmetic verification of those representations;
6. the software version, compiler, hardware and runtime;
7. a checksum of the code and data files.

For a theorem-level computational claim, the code and data are part of the mathematical certificate.

11.4 Independent checks

Three checks are particularly important. First, direct sparse-set enumeration should agree with the bitset algorithm for small B . Second, every displayed representation should be evaluated independently. Third, the same extremal values should be recovered from a separate implementation using a different data structure.

These checks do not prove an asymptotic conjecture, but they do isolate computational error from mathematical uncertainty.

12 Computational Observations for the Diagonal Model

The following values are reported for the finite-height model $H = X = B$. They are not claims about the unrestricted word metric.

Table 1: Reported exact diagonal values $M_k(B, B)$ for selected palettes and bounds.

B	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k = 10$
10^2	3	3	2	2	2	2
10^3	4	4	4	4	3	3
10^4	5	5	5	5	4	4
10^5	6	6	5	5	5	5
10^6	7	7	6	6	6	5
10^7	8	7	7	6	6	6

For the palette $S_4 = \{2, 3, 5, 7\}$ the reported diagonal values are

$$M_4(10^3, 10^3) = 4, \quad M_4(10^4, 10^4) = 5, \quad M_4(10^6, 10^6) = 6, \quad M_4(10^7, 10^7) = 7.$$

The data show a steady increase in the worst-case term count over the displayed range, together with substantial gains from adding bases. They do not, however, distinguish among all slowly growing asymptotic laws.

12.1 Record primes for S_4

For $k = 4$ and $B = 10^7$, the reported first primes requiring successive observed lengths are:

Table 2: Reported record primes for $S_4 = \{2, 3, 5, 7\}$ in the diagonal finite-height model at $B = 10^7$.

Length	Prime	One shortest representation reported
1	2	2
2	11	$-2^4 + 3^3$
3	107	$-2^{11} - 2^5 + 3^7$
4	1123	$-7^5 - 2^{10} - 3^6 + 3^9$
5	11717	$-2^{19} + 5^3 + 7^3 + 2^{12} + 3^{12}$
6	607819	$-5^{10} - 2^7 + 2^4 + 5^8 + 3^{13} + 2^{23}$
7	7100879	$-5^{10} - 7^4 - 2^7 + 2^{15} + 3^{10} + 2^{23} + 2^{23}$

The last representation uses the repeated atom 2^{23} twice. In coefficient notation its contribution is $2 \cdot 2^{23}$ and therefore costs two atomic terms.

The two larger identities are

$$607819 = -5^{10} - 2^7 + 2^4 + 5^8 + 3^{13} + 2^{23},$$

and

$$7100879 = -5^{10} - 7^4 - 2^7 + 2^{15} + 3^{10} + 2^{23} + 2^{23}.$$

The latter expands numerically as

$$7100879 = -9765625 - 2401 - 128 + 32768 + 59049 + 8388608 + 8388608.$$

12.2 Normalized scale comparison

Set

$$L(B) = \frac{\log B}{\log \log B}.$$

At $B = 10^2, 10^3, \dots, 10^7$, the values of $L(B)$ are approximately

$$3.02, 3.57, 4.15, 4.71, 5.26, 5.80.$$

For $k = 4$, the reported maxima

$$2, 4, 5, 5, 6, 7$$

are numerically on the same broad scale, but this agreement is only qualitative. In particular, the ratio between the observed maximum and $L(B)$ changes noticeably across the small range, and six points are far too few to support an asymptotic constant.

13 Prime Targets Versus All Integers

The all-integer quantity

$$N_k(X, H) = \max_{1 \leq n \leq X} \lambda_{k,H}(n)$$

provides an important control experiment. By definition,

$$M_k(X, H) \leq N_k(X, H).$$

The elementary counting lower bound has the same leading order for primes and all integers because

$$\pi(X) \sim \frac{X}{\log X}$$

and therefore

$$\log \pi(X) = \log X - \log \log X + o(\log X).$$

Thus prime sparsity does not change the first-order entropy threshold.

What remains genuinely open is whether arithmetic structure produces a systematic difference between the extremal prime and all-integer populations. There are several conceivable outcomes:

1. $M_k(X, X)$ and $N_k(X, X)$ have the same order and comparable constants;
2. they have the same order but different leading constants;
3. primes are sufficiently atypical that one of the two extremal laws has a different lower-order fluctuation pattern.

The second and third possibilities can be investigated computationally before any asymptotic theorem is available.

14 Hard Primes and Representation Profiles

A prime p is called a *hard prime* for (k, X, H) if it attains the maximum:

$$\lambda_{k,H}(p) = M_k(X, H).$$

The set of hard primes is more informative than a single record value. It can be studied through:

size of p , $p \bmod q$, number of shortest representations, base frequencies, largest exponent used.

For a shortest representation, define the base-profile vector

$$(w_2, w_3, \dots, w_{p_k}),$$

where w_p counts the number of signed atoms based at p . One can also record the maximum exponent

$$E_{\max} = \max\{a : p^a \text{ occurs in the chosen representation}\}.$$

These statistics may help distinguish a genuinely combinatorial coverage phenomenon from one driven by a small number of special algebraic identities.

15 Height Inflation and the Unrestricted Problem

The two-parameter model permits a precise formulation of the relationship between bounded and unrestricted representations.

Define the *stabilization height* of a target n by

$$H_k^*(n) = \min\{H \geq 2 : \lambda_{k,H}(n) = \lambda_k(n)\}.$$

This exists because an unrestricted shortest representation uses finitely many atoms.

At the range level define

$$H_k^*(X) = \min\{H \geq 2 : M_k(X, H) = M_k^\infty(X)\}.$$

Studying $H_k^*(X)$ would quantify how much larger than the target range one ever needs to go to realize unrestricted shortest prime representations.

A particularly useful intermediate problem is therefore:

Question 15.1. *Is there a polynomial height exponent $\alpha_k < \infty$ such that for all sufficiently large X , every prime $p \leq X$ has an unrestricted shortest representation using only atoms at most X^{α_k} ?*

Any such height-control theorem would transfer bounded-height information toward the unrestricted setting. Conversely, evidence that $H_k^*(X)$ grows faster than every power of X would demonstrate that unrestricted cancellation is an essential part of the problem rather than a technical complication.

16 Proof Strategies and Concrete Intermediate Problems

16.1 Greedy approximation by pure powers

A natural approach is to approximate the current remainder by an admissible pure power. For one base, the largest available power reduces the remainder only by a constant proportion in each step, leading to a logarithmic number of steps. With several bases, the union of the geometric sequences is less rigid.

A useful quantitative target would be an approximation statement of the form

$$\forall n \text{ sufficiently large, } \exists a \in A_k(n^C) \text{ with } |n - a| \leq \frac{n}{(\log n)^\delta}$$

for some fixed $C, \delta > 0$. Such an estimate would not immediately imply the main conjecture, but it would provide a concrete mechanism for a sublogarithmic greedy algorithm.

At present this remains speculative. The available pure-power set is much thinner than the set of mixed power products, so one cannot simply import discrepancy results from the denser model.

16.2 Iterated sumset growth

A second route is additive combinatorics. One seeks lower bounds for

$$|\Sigma_{k,H}(h) \cap [1, X]|$$

and, more strongly, for the number of residue classes occupied modulo carefully chosen moduli.

The conjecture suggests that the threshold

$$h = C_k \frac{\log X}{\log \log X}$$

should already produce enough expansion to capture all primes. A density theorem such as

$$\frac{|\Sigma_{k,X}(h) \cap [1, X] \cap \mathbb{P}|}{\pi(X)} \rightarrow 1$$

would be a meaningful first milestone.

16.3 Congruence and covering arguments

Congruence arguments may serve two complementary purposes. First, they can establish lower bounds by identifying residue classes inaccessible to short sums. Second, they can certify computational nonexistence without enumerating every formal representation.

The double-base literature illustrates how modular information can sharply reduce the search space. Developing analogous congruence certificates for the pure-power dictionary is a promising direction.

16.4 Exponential Diophantine methods

Equations of the form

$$\pm p_i^{a_i} \pm p_j^{a_j} = n$$

and higher-term analogues are exponential Diophantine equations. The fixed-term S -unit theory gives strong sparsity statements, but the main conjecture requires a term budget growing with X . The challenge is therefore to combine structural Diophantine information with a growing-complexity regime.

16.5 Prime-specific methods

The prime-target restriction creates opportunities as well as difficulties. A density-one result for integers does not automatically imply a corresponding result for primes, but analytic information about primes in residue classes may help once suitable local distribution estimates for the signed sumsets are available.

17 Further Conjectures and Testable Questions

17.1 Prime versus integer extremality

Conjecture 17.1. *For every fixed $k \geq 2$,*

$$M_k(B, B) = \Theta(N_k(B, B)) \quad (B \rightarrow \infty).$$

This is deliberately weaker than the Pulickal Conjecture if the all-integer quantity is easier to control. Its computational verification over increasing ranges would be useful even before a proof is available.

17.2 A density-one version

Conjecture 17.2. *For each fixed $k \geq 2$ there exists $C_k > 0$ such that, with*

$$h_B = C_k \frac{\log B}{\log \log B},$$

one has

$$\frac{|\{p \leq B : p \text{ prime, } \lambda_{k,B}(p) \leq h_B\}|}{\pi(B)} \rightarrow 1.$$

This is weaker than the extremal statement because it allows a thin exceptional set of hard primes.

17.3 Leading constants

Question 17.3. *Does the limit*

$$\lim_{B \rightarrow \infty} \frac{M_k(B, B) \log \log B}{\log B}$$

exist for any fixed $k \geq 2$? If so, how does it depend on k ?

The counting argument gives a universal lower bound of 1 for the liminf, but nothing here suggests that 1 should be the actual limit.

17.4 Base dependence

The sequence

$$M_1(B, B), M_2(B, B), M_3(B, B), \dots$$

should be studied for fixed B as well as asymptotically. Numerically, the first additional bases can produce substantial finite-range reductions, while later additions show diminishing returns. A theoretical understanding of this transition would clarify how much of the conjectured phenomenon comes from the first two bases and how much comes from palette growth.

17.5 Height regimes

A natural phase diagram uses

$$H = X^\alpha$$

with $\alpha > 0$, or more generally

$$\log H \asymp (\log X)^\beta.$$

The diagonal corresponds to $\alpha = 1$. The counting lower bound persists across every fixed polynomial height regime. It would be useful to determine whether the conjectural upper bound also persists uniformly in α .

18 Limitations

The present paper has four explicit limitations.

First, the principal asymptotic upper bound is conjectural. The main theorem-level contribution is the counting lower bound, not a proof of the conjecture.

Second, the finite computations reported here are evidence for a finite model. They do not establish anything directly about unrestricted representations with arbitrarily large cancelling atoms.

Third, the numerical range, while substantial for computation, is tiny on the scale of asymptotic analysis. Values through 10^7 cannot distinguish $\log B / \log \log B$ from many competing slowly growing functions.

Fourth, the broader representation theory is not new. The intended contribution is the focused prime-target, finite-height extremal formulation and its associated computational programme. The literature comparison should be understood accordingly.

19 Recommended Reproducibility Standard

A publication version of this project should package the computational component as a self-contained supplement containing:

1. a source-controlled implementation;
2. a machine-readable table of $\lambda_{k,H}(p)$ for all computed primes;
3. exact shortest representations for all hard primes;
4. independent small-range verification;
5. deterministic build instructions;
6. code and data checksums;
7. explicit hardware and runtime information.

For the 10^7 computation, the supplement should make it possible for another researcher to reproduce both the final maximum and the entire set of record primes. This is preferable to publishing only a handful of attractive representations.

20 Discussion

The motivating phenomenon can now be stated without ambiguity. A fixed finite palette of prime bases produces only logarithmically many pure-power atoms up to height B . Consequently, no bounded number of atoms can cover all primes indefinitely, and in fact the counting obstruction forces at least

$$(1 - o(1)) \frac{\log B}{\log \log B}$$

terms in the worst case. At the other end, signed binary arithmetic provides a logarithmic upper bound.

The resulting gap is not a vague request for “better estimates”. It is a specific structural question about whether a union of a few geometric progressions, under signed addition and atomic multiplicity cost, exhibits enough additive expansion to reach every prime target at the counting threshold.

The two-parameter model sharpens this statement further. It distinguishes target size from permitted atom height and creates a natural bridge between finite computation and the unrestricted word metric. The diagonal model is the cleanest finite problem; the height-inflation function $H_k^*(X)$ is a natural tool for understanding how far that finite problem is from the unrestricted one.

The finite data show that, for the first several million integers and several small prime palettes, the required term counts are surprisingly modest. That observation is compatible with the conjecture, but it should not be oversold. The significance of the data is that they provide a concrete experimental landscape in which one can search for record primes, identify local obstructions, compare primes with generic integers, and test candidate proof mechanisms.

The main research programme therefore has three logically distinct tracks:

1. **structure:** prove new theorems for pure-power sumsets or derive upper bounds for growing term budgets;
2. **computation:** extend the exact diagonal tables, characterize hard primes, and map the dependence on k and H ;
3. **transfer:** understand the relation between finite-height extrema and the unrestricted word metric.

A positive solution of the Pulickal Conjecture would establish the same order of growth already known in denser multibase systems, but in a substantially thinner pure-power dictionary and under a prime-target extremal requirement. A negative solution would be equally informative: it would show that the counting threshold is not automatically sufficient once mixed power products are removed, thereby identifying a genuine obstruction to sparse additive coverage.

21 Conclusion

We have formulated a finite and exact two-parameter representation problem for signed pure powers of a fixed finite set of prime bases. The diagonal prime extremal function

$$M_k(B, B) = \max_{\substack{p \leq B \\ p \text{ prime}}} \lambda_{k,B}(p)$$

is the principal object.

Three facts are unconditional. First, the atom set has size $\asymp_k \log B$. Second, a counting argument forces

$$M_k(B, B) \geq (1 - o(1)) \frac{\log B}{\log \log B}.$$

Third, signed-binary expansion gives

$$M_k(B, B) \leq \frac{\log B}{2 \log 2} + O(1).$$

The Pulickal Conjecture proposes that for every fixed $k \geq 2$ the lower-bound scale is sharp:

$$M_k(B, B) = \Theta_k\left(\frac{\log B}{\log \log B}\right).$$

The conjecture is a statement about the finite-height pure-power system itself; it is not a disguised assertion about power products, and it does not follow from existing multibase Hamming-weight theorems.

The computational component supports the idea that the system has unusually economical finite representations, but the asymptotic question remains open. The most useful next advances would be a nontrivial improvement on the present $O(\log B)$ upper bound, a density-one theorem at the conjectured scale, a sharp theory of hard primes, or a height-control theorem connecting finite-height and unrestricted shortest representations.

The proposed problem is therefore deliberately narrow but mathematically substantive: it isolates a sparse additive dictionary, a prime-target extremal statistic, a universal counting barrier, and a specific conjectural upper bound. The central question is not whether primes can occasionally be written with a few small prime powers—they clearly can—but whether, for a fixed multi-prime palette, the worst-case complexity grows at essentially the smallest order allowed by counting.

A Coefficient-Cost Geometry

For fixed (k, H) let

$$\mathcal{D}_{k,H} = \{a \in A_k(H)\}.$$

A representation of n corresponds to an integer coefficient vector

$$c = (c_a)_{a \in \mathcal{D}_{k,H}} \in \mathbb{Z}^{|A_k(H)|}$$

satisfying

$$\sum_{a \in A_k(H)} c_a a = n.$$

The cost is the ℓ^1 norm

$$\|c\|_1 = \sum_{a \in A_k(H)} |c_a|.$$

Thus $\lambda_{k,H}(n)$ is the minimum ℓ^1 norm of an integer point on the affine hyperplane

$$\sum_a c_a a = n.$$

This lattice viewpoint clarifies why repeated atoms are charged linearly and why arbitrary coefficients are not free.

If one records the coefficient vector rather than merely the value of the optimum, one obtains additional statistics: support size, signed support, coefficient multiplicities, base distribution, and maximum exponent. These can be useful in experimental attempts to identify hidden structure.

B Exact Layer Recurrence

Let

$$E_h(k, H) = \left\{ \sum_{j=1}^h \varepsilon_j a_j : \varepsilon_j \in \{-1, +1\}, a_j \in A_k(H) \right\}.$$

Then

$$E_0(k, H) = \{0\}$$

and

$$E_{h+1}(k, H) = E_h(k, H) + (\pm A_k(H)).$$

Every element of $E_h(k, H)$ lies in $[-hH, hH]$.

For every target $n \geq 1$,

$$\lambda_{k,H}(n) = \min\{h : n \in E_h(k, H)\}.$$

This equality follows directly from the definition. The dynamic programme is therefore an exact shortest-path computation in a layered Cayley graph of $\mathbb{Z}$ generated by $\pm A_k(H)$.

C Reference Pseudocode

INPUT: target bound X, height bound H, palette size k

1. Construct the distinct atom set

$$A = \{1\} \text{ union } \{p_i^a : 1 \leq i \leq k, a \geq 1\}.$$

2. Enumerate all primes $p \leq X$.

Set $\text{lambda}[p] := \text{UNSET}$ for each such prime.

3. Initialize $E := \{0\}$.

4. For $h = 1, 2, 3, \dots$:

$E_{\text{new}} := \text{empty set}$

 for each a in A :

$E_{\text{new}} := E_{\text{new}} \text{ union } (E + a) \text{ union } (E - a)$

$E := E_{\text{new}}$

 for each prime $p \leq X$ with $\text{lambda}[p] \text{ UNSET}$:

 if p in E :

```

lambda[p] := h

if every prime has been assigned:
    return max(lambda[p])

```

For an implementation using bitsets, the abstract set shifts become word-parallel shifts and bitwise OR operations. The bitset must cover the full interval $[-hH, hH]$ at layer h ; truncating this interval can silently destroy valid future representations.

D A Minimal Verification Checklist

For a computational release associated with this paper, the following statements should all be mechanically verifiable:

1. the atom list is complete and contains no accidental duplicates;
2. every displayed representation evaluates to its stated target;
3. every reported length is attained by an explicit representation;
4. no shorter representation exists according to the exact dynamic programme;
5. an independent implementation agrees on small and medium instances;
6. the final maximum agrees with the maximum of the stored per-prime lengths.

E Reported Record Data

For convenience, the reported S_4 record sequence at diagonal height $B = 10^7$ is

2, 11, 107, 1123, 11717, 607819, 7100879,

corresponding to lengths

1, 2, 3, 4, 5, 6, 7.

The entries should be understood as finite-height records. They are not being asserted to be records for the unrestricted problem.

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